System Verilog Projects

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Project-1: CRC-8 Generator Design and Verification using SystemVerilog

Project Description:

Designed and verified a CRC-8 (Cyclic Redundancy Check) generator using SystemVerilog for error detection in digital communication. Implemented the CRC algorithm using polynomial 8'h07 ($x^8 + x^2 + x + 1$) with an initial CRC value of 8'h00. Developed a class-based verification environment using SystemVerilog OOP concepts, including Transaction, Generator, Driver, Monitor, Scoreboard, Environment, and Test components. Verified the design functionality through simulation and scoreboard-based checking.

Key Features:

- CRC-8 generation using polynomial 8'h07.
- Initial CRC value of 8'h00.
- Class-based SystemVerilog verification environment.
- Generator, Driver, Monitor and Scoreboard components.
- Functional verification using simulation.

Tools Used: SystemVerilog, EDA Playground.

Verification: Class-based testbench and scoreboard-based functional verification.

EDA Link: https://www.edaplayground.com/x/i7Np

Project-2: Traffic Light Controller Design and Verification using SystemVerilog

Project Description:

Designed and verified an FSM-based Traffic Light Controller using SystemVerilog. Implemented five operating states: RED, GREEN, YELLOW, PEDESTRIAN, and EMERGENCY. Developed a traffic light control mechanism with configurable signal durations, pedestrian request handling, and emergency vehicle priority. Created a class-based verification environment with transaction, generator, driver, monitor, agent, environment, and scoreboard components. Implemented an independent reference FSM in the scoreboard to verify DUT outputs and state transitions.

Key Features:

- Five-state FSM-based traffic control.
- Pedestrian request handling.
- Emergency vehicle handling.
- Configurable traffic signal durations.
- Independent reference FSM scoreboard.

Tools Used: SystemVerilog, EDA Playground.

Verification: Class-based testbench and scoreboard-based functional verification.

EDA Link: https://www.edaplayground.com/x/h7_w

Project-3: 8b/10b Encoder and Decoder Design and Verification using SystemVerilog

Project Description:

Designed and verified an 8b/10b Encoder and Decoder using SystemVerilog. Implemented the encoding logic to convert 8-bit input data into 10-bit encoded symbols and decoding logic to recover the original 8-bit data. Developed a testbench to verify encoding and decoding functionality and ensure correct data recovery through simulation.

Key Features:

- 8-bit to 10-bit data encoding.
- 10-bit to 8-bit data decoding.
- Encoder and decoder functionality verification.
- Simulation-based functional testing.

Tools Used: SystemVerilog, EDA Playground.

Verification: Testbench-based functional verification.

EDA Link: **https://www.edaplayground.com/x/SgFa**

Project-4: Smart Vending Machine Controller using SystemVerilog

Project Description:

Designed and implemented a Smart Vending Machine Controller using SystemVerilog based on a Moore Finite State Machine (FSM). Implemented seven FSM states: IDLE, ACCEPT, CHECK, DISPENSE, CHANGE, REFUND, and ERROR to control vending machine operations. The design supports coin acceptance (₹5, ₹10, and ₹20), product selection (Water, Juice, and Snacks), balance management, automatic product dispensing, change calculation, cancellation, and refund handling. Verified the controller functionality using a class-based SystemVerilog testbench with an independent reference FSM scoreboard.

Key Features:

- Seven-state Moore FSM architecture.
- Coin acceptance and balance management.
- Three product selections with different prices.
- Automatic product dispensing.
- Change calculation and refund handling.
- Cancellation and invalid product selection handling.
- Independent reference FSM for scoreboard checking.

Tools Used: SystemVerilog, EDA Playground.

Verification: Class-based testbench and scoreboard-based functional verification.

EDA Link: https://www.edaplayground.com/x/gC2m